

IMPORTANT NOTE

- 1) This sample paper serves solely as a visual representation of the question paper.
- 2) Each question in the question paper is mapped with corresponding Course Outcome(s) (CO) and Revised Bloom's Taxonomy (RBT) Level. However the question-wise CO-RBT mapping given below is indicative only and will vary in the final question paper.
- 3) Students should refer to the latest course syllabus and question paper pattern for exam preparation.
- 4) The final version of the End Term Examination question paper may differ in terms of layout, design, and content. This sample paper should be viewed for reference purpose only and not as final/official question paper.

24251CSE58040

Paper Code: A

Course Code: CSE316

Course Title: OPERATING SYSTEMS

Time Allowed: 03:00hrs.

Max Marks: 60

Read the following instructions carefully before attempting the question paper.

1. Match the Paper Code shaded on the OMR Sheet with the Paper code mentioned on the question paper and ensure that both are the same.
2. This question paper contains 60 questions of 1 mark each. 0.25 marks will be deducted for each wrong answer.
3. Attempt all the questions in serial order.
4. Do not write or mark anything on the question paper and/or on rough sheet(s) which could be helpful to any student in copying, except your registration number on the designated space.
5. Submit the question paper and the rough sheet(s) along with the OMR sheet to the invigilator before leaving the examination hall.

Q(1) In a deadlock situation, which condition ensures that resources are held by processes and cannot be taken away?

- | | |
|----------------------|-------------------|
| (a) Mutual Exclusion | (b) Hold and Wait |
| (c) No Preemption | (d) Circular Wait |

CO4,L3

Q(2) Which of the following best describes the method of sequential file access?

- (a) read bytes one at a time, in order
- (b) random access according to the given byte number
- (c) read/write randomly by record
- (d) read/write sequentially by record

CO1,L6

Q(3) Which of the following would best explain why the LRU page replacement algorithm generally performs better than FIFO in most practical situations?

- (a) LRU keeps track of the entire history of page references.
- (b) LRU assumes that pages used recently are more likely to be used again soon.
- (c) FIFO only works well with a specific sequence of page references.
- (d) LRU is faster in terms of time complexity compared to FIFO.

CO3,L4

Q(4) For a process P running on a CPU, which of the following events will consistently cause the operating system to perform a context switch, forcing process P to transition to a non-running state (such as ready or blocked)?

- (a) P makes a blocking system call to read a block of data from the disk.
- (b) None of the option is correct
- (c) An interrupt is raised by the disk to deliver data requested by some other process.
- (d) A timer interrupt is raised by the hardware

CO2,L2

Q(5) Which scheduling algorithm selects the process with the shortest estimated run-time for the next execution?

- | | |
|------------------------------|-----------------------------------|
| (a) Round Robin | (b) Priority Scheduling |
| (c) Shortest Job First (SJF) | (d) First Come First Serve (FCFS) |

CO2,L4

Q(6) Which will be the first process to start using SRTF algorithm

Process	Burst Time	Arrival time
P1	4	3
P2	5	1
P3	3	3
P4	2	2

- | | |
|--------|--------|
| (a) P1 | (b) P2 |
| (c) P3 | (d) P4 |

Q(7) Dissect What is file sharing?

- (a) Sharing of files between multiple users
- (b) Sharing of devices between multiple users
- (c) Sharing of memory between multiple processes
- (d) Sharing of CPU between multiple users

CO6,L4

Q(8) Choose Which of the following is an example of starvation?

- (a) A process waiting for a resource that is held by another process.
- (b) A process waiting for a resource that is not available.
- (c) A process waiting for a resource that is requested by another process but never granted.
- (d) A process holding a resource that is requested by another process.

CO4,L3

Q(9) What are the two kind of semaphores?

- (a) Mutex and counting
- (b) Binary and counting
- (c) binary and decimal
- (d) Counting and Mutex

CO2,L1

Q(10) Each process has a segment of code, called a _____, in which the process may be changing common variables, updating a table, writing a file, and so on

- (a) None-critical section
- (b) Critical section
- (c) Exit Section
- (d) Entry Section

CO3,L3

Q(11) Define the primary function of an operating system.

- (a) Run user applications
- (b) Compile programs
- (c) Manage hardware resources
- (d) Design software

CO1,L1

Q(12) If a system employs a round-robin scheduling algorithm with a time quantum of 2 milliseconds. If there are 60 process and each process burst time is 5 milliseconds to execute, how many processes can be completed in 250 milliseconds?

- (a) 50
- (b) 25
- (c) 10
- (d) None of these

CO2,L4

Q(13) Which synchronization mechanism is most prone to "busy waiting"?

- (a) Mutex
- (b) Semaphore
- (c) Monitor
- (d) Spinlock

CO3,L4

Q(14) Relate Which of the following conditions makes a system unsafe but not deadlocked?

- (a) Mutual Exclusion
- (b) No Preemption
- (c) Circular Wait
- (d) Unsafe Resource Allocation

CO4,L3

Q(15) A process stack does not contain _____

- (a) Function parameters
- (b) Local variables
- (c) Return addresses
- (d) PID of child process

CO1,L3

Q(16) What is paging in memory management?

- (a) A technique to increase CPU speed
- (b) A method to divide memory into fixed-size blocks
- (c) A way to optimize disk scheduling
- (d) A method to avoid memory leaks

CO4,L6

Q(17) Simplify Given the following pseudo-code, what will be the outcome if two processes execute simultaneously?

```
lock(mutex)
counter = counter + 1
unlock(mutex)
```

- (a) The counter will always be incremented correctly
- (b) Race conditions will occur
- (c) The counter will remain unchanged
- (d) Deadlock will occur

CO3,L4

Q(18) Which component of an operating system is responsible for managing system calls?

- (a) Process Scheduler
- (b) File Manager
- (c) Kernel
- (d) Device Driver

CO1,L1

Registration No.: _____

Q(19) Which scheduling algorithm assigns priorities to processes based on their characteristics, such as age, memory requirements, or I/O usage?

- (a) Round Robin
- (b) Shortest Job First (SJF)
- (c) Priority Scheduling
- (d) Multi-Level Feedback Queue

CO2,L4

Q(20) Which of the following is a method to securely maintain passwords?

- (a) Using plaintext storage
- (b) Hashing passwords before storage
- (c) Sending passwords over unencrypted channels
- (d) Reusing passwords across multiple systems

CO4,L2

Q(21) Which scheduling algorithm may suffer from "starvation," where a low-priority process never gets CPU time?

- (a) Priority Scheduling
- (b) Round Robin
- (c) Shortest Job First (SJF)
- (d) First-Come, First-Served (FCFS)

CO1,L4

Q(22) Scheduler activations are a technique used to

- (a) Allocate resources to threads
- (b) Prevent deadlocks
- (c) Switch between processes
- (d) Schedule threads within a process

CO3,L3

Q(23) If a semaphore S is initialized to 3 and a process performs three wait(S) operations in sequence, what will be the final value of S?

- (a) 0
- (b) -3
- (c) 3
- (d) -3 or a blocked state

CO3,L1

Q(24) In the sequential access method, information in the file is processed _____

- (a) one disk after the other, record access doesn't matter
- (b) one record after the other
- (c) one text document after the other
- (d) none of the mentioned

CO6,L2

Q(25) Which synchronization problem involves multiple readers and writers accessing a shared resource?

- (a) Producer-Consumer Problem
- (b) Dining Philosophers Problem
- (c) Reader-Writer Problem
- (d) Critical Section Problem

CO6,L3

Q(26) Distinguish/recognize the best example for spooled devices

- (a) A line printer is used to print the output of the number of jobs
- (b) A terminal used to enter input data to a running program
- (c) A secondary storage device in virtual memory system
- (d) A graphic display device

,L4

Q(27) Determine the following is not a method for preventing deadlocks?

- (a) Deadlock avoidance
- (b) Deadlock detection
- (c) Deadlock recovery
- (d) Deadlock termination

CO4,L3

Q(28) Which disk scheduling algorithm would you design to ensure a balance between speed and fairness in file retrieval?

- (a) Create an SSTF-based algorithm with round-robin enhancements.
- (b) Use a simple FCFS strategy.
- (c) Design a LIFO-only system.
- (d) Combine FIFO with LIFO for flexibility.

CO6,L4

Q(29) A computer uses 46-bit virtual address, 32-bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level table (T1), which occupies exactly one page. Each entry of T1 stores the base address of a page of the second-level table (T2). Each entry of T2 stores the base address of a page of the third-level table (T3). Each entry of T3 stores a page table entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16 way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes. What is the size of a page in KB in this computer?

- (a) 2
- (b) 4
- (c) 16
- (d) 8

,L6

Q(30) What does the access matrix represent?

- (a) Rows-Domains, Columns-Objects
- (b) Rows-Objects, Columns-Domains
- (c) Rows-access list, Columns-Domains
- (d) Rows-Domains, Columns-access list

CO4,L6

Registration No.: _____

Q(31) In paged memory systems, if the page size is increased, then the internal fragmentation generally

- (a) becomes less
- (b) becomes more
- (c) remains constant
- (d) none of the mentioned

CO1,L6

Q(32) A disk scheduling algorithm uses the SCAN method. The disk has 200 cylinders, numbered 0 to 199. The head is currently at cylinder 70 and moving towards 0. The queue of requests is as follows: 50, 180, 40, 160, 10. What is the total head movement using the SCAN algorithm?

- (a) 190 cylinders
- (b) 170 cylinders
- (c) 230 cylinders
- (d) 210 cylinders

CO5,L3

Q(33) Which of the following correctly describes the difference between a safe and an unsafe state?

- (a) A safe state guarantees no deadlock will occur, whereas an unsafe state might lead to a deadlock.
- (b) A safe state indicates that deadlock has occurred, whereas an unsafe state guarantees deadlock-free operation.
- (c) A safe state means resources are allocated optimally, while an unsafe state means resources are over-allocated.
- (d) A safe state means resources are allocated optimally, while an unsafe state means resources are over-allocated.

CO5,L2

Q(34) Which disk scheduling algorithm selects the request that is closest to the current disk head position and minimizes seek time?

- (a) First-Come-First-Serve (FCFS)
- (b) Shortest Seek Time First (SSTF)
- (c) Circular LOOK (C-LOOK)
- (d) SCAN

CO5,L3

Q(35) Interpret for a given precedence graph in a concurrent system, how would you identify potential deadlock conditions?

- (a) By locating cycles in the graph that represent circular dependencies.
- (b) By checking for paths that lead from start to finish.
- (c) By analyzing whether any node in the graph has multiple incoming edges.
- (d) By determining if any process has completed all its tasks.

CO3,L3

Q(36) What are the 2 views of an Operating system

- (a) Exception view and Trap view
- (b) Resource view and Control view
- (c) User view and System view
- (d) Interrupt view and System call view

,L1

Q(37) In paging, the page table contains the mapping between which of the following?

- (a) Logical addresses and segment numbers
- (b) Logical addresses and physical addresses
- (c) Logical page numbers and physical frame numbers
- (d) Physical addresses and logical page numbers

CO5,L4

Q(38) Determine correct application of semaphores in concurrent programming?

- (a) Scheduling the execution of background tasks
- (b) Allowing multiple processes to read from a shared file simultaneously
- (c) Managing user input in a graphical user interface
- (d) Ensuring that only one process can write to a file at a time

CO5,L3

Q(39) Separate out the false statement from following for CPU scheduling

- (a) State change from Running to Ready is Preemptive type
- (b) State change from Waiting to Ready is Preemptive type
- (c) State change from Running to Waiting is Non-preemptive type
- (d) State change from Waiting to Ready is Non-preemptive type

,L4

Q(40) By increasing the number of frames to the process, number of page faults should decrease but instead they are increasing this problem is called?

- (a) Bells Anomaly
- (b) Belady's Anomaly
- (c) Peterson's Anomaly
- (d) Alen Anomaly

,L3

Q(41) Determining What is the Memory Management Unit's (MMU) purpose?

- (a) To manage memory hierarchy
- (b) To translate virtual addresses to physical addresses
- (c) To allocate memory resources to processes
- (d) To control access to memory

CO5,L2

Registration No.: _____

Q(42) If semaphore value is negative:

- (a) its magnitude is the number of processes waiting on that semaphore or blocked
- (b) it is invalid
- (c) No operation can be further performed on it until the signal operation is performed on it
- (d) None of the mentioned

,L2

Q(43) What is the completion time of process 1, process 2 and process 3 using Round Robin algorithm with Time quantum equal to 2?

- (a) 1, 2, 9 and 8 respectively
- (b) 1, 7 and 8 respectively
- (c) 2, 9 and 10 respectively
- (d) 2, 6 and 9 respectively.

CO1,L4

Q(44) Analyze the following scenario: A system using the Optimal page replacement algorithm achieves a lower page fault rate compared to LRU and FIFO under certain conditions. What is the primary reason behind this?

- (a) Optimal replaces the page that will not be used for the longest period in the future.
- (b) Optimal algorithm maintains a list of all pages in the memory.
- (c) Optimal algorithm relies on past page usage to predict future requests.
- (d) Optimal algorithm is specifically designed to minimize computational overhead.

CO3,L4

Q(45) Analyze which of the following is a non-preemptive scheduling algorithm?

- (a) Round Robin
- (b) Shortest Remaining Time First
- (c) First Come First Serve
- (d) Priority Scheduling

CO3,L4

Q(46) Which stage of a process does not belong to primary memory:-

- (a) New
- (b) Reddy
- (c) Running
- (d) All of these

CO1,L3

Q(47) Which of the following problems can be solved using Peterson's solution?

- (a) Deadlock
- (b) Process Synchronization for two processes
- (c) Page Replacement
- (d) CPU Scheduling

CO3,L3

Q(48) What is the purpose of the popen function in IPC?

- (a) To open a file for reading
- (b) To create a pipe and execute a command
- (c) To close a pipe and terminate a process
- (d) To create a new process

CO6,L2

Q(49) Plan a device management strategy that efficiently manages dedicated, shared, and virtual devices in a cloud environment. What key feature would you implement?

- (a) Static resource partitioning
- (b) Dynamic resource allocation
- (c) Device prioritization
- (d) Load balancing

CO6,L6

Q(50) If the process can be moved during its execution from one memory segment to another, then binding must be

- (a) Delayed until run time
- (b) Preponed to compile time
- (c) Preponed to load time
- (d) None of the mentioned

CO1,L6

Q(51) Which of the following scenarios best illustrates the occurrence of internal fragmentation in an operating system?

- (a) A process is allocated more memory than it requested, leaving unused space within the allocated block.
- (b) A process requests memory, but it cannot be allocated due to the scattered nature of available free memory blocks.
- (c) Memory is divided into fixed-sized blocks, and a process fits perfectly into one of these blocks.
- (d) The total available memory is sufficient for a process, but no single contiguous block is large enough to accommodate it.

CO3,L4

- Q(52) Which of the following best describes a process life cycle in an OS?
(a) The sequence of actions taken by a user to start and stop a process.
(b) The series of stages a process goes through from creation to termination.
(c) The order in which processes are executed in a multiprogramming environment.
(d) The method by which processes are synchronized in a multiprocessing system.
CO1,L1
- Q(53) Which of the following is not a necessary condition for deadlock to occur?
(a) Mutual Exclusion
(b) Hold and Wait
(c) Preemption
(d) Circular Wait
CO4,L3
- Q(54) What information does the Process Control Block (PCB) contain: -
(a) Process ID
(b) State
(c) Priority
(d) All of these
CO1,L2
- Q(55) What is a system call in the context of an OS?
(a) A function that enables the OS to call external software.
(b) A command-line instruction for managing processes.
(c) A programmatic way for a process to request a service from the OS.
(d) A method for compiling code in the OS environment.
CO1,L2
- Q(56) Mutual exclusion implies that :
(a) if a process is executing in its critical section, then no other process must be executing in their critical sections
(b) if a process is executing in its critical section, then other processes must be executing in their critical sections
(c) if a process is executing in its critical section, then all the resources of the system must be blocked until it finishes execution
(d) none of the mentioned
CO3,L3
- Q(57) Which of the following is NOT a solution to the critical section problem?
(a) Monitors
(b) Semaphores
(c) Paging
(d) Locks
CO4,L3
- Q(58) Every process in an operating system has its own _____.
(a) address space and global variables
(b) pending alarms, signals, and signal handlers
(c) open files
(d) all of the mentioned
CO1,L3
- Q(59) In a system that does not support swapping _____
(a) the compiler normally binds symbolic addresses (variables) to relocatable addresses
(b) the compiler normally binds symbolic addresses to physical addresses
(c) the loader binds relocatable addresses to physical addresses
(d) binding of symbolic addresses to physical addresses normally takes place during execution
CO5,L4
- Q(60) What is the name of the component responsible for task switching in multitasking environments?
(a) Scheduler
(b) Dispatcher
(c) Interrupt Handler
(d) Process Table
CO2,L4

--End of Question paper--